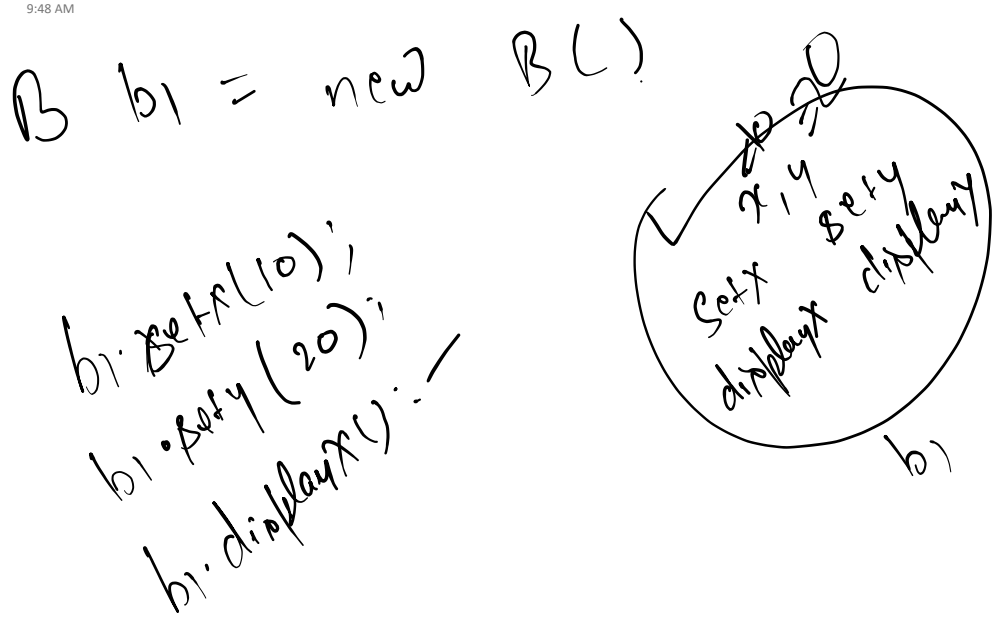
CS Session 3

Monday, 16 February 2026 9:48 AM



Constructors in Java

A **constructor** in Java is a special member method that is automatically called when an object of a class is created. It is used to **initialize the object**.

Key Characteristics of Constructors

1. **Same name as class**
   The constructor name must be exactly the same as the class name.
2. **No return type**
   Constructors do not have any return type, not even void.
3. **Called automatically**
   It is invoked automatically when an object is created using the new keyword.
4. **Used to initialize object variables**
   Constructors help assign initial values to instance variables.

Syntax

```
class ClassName {
   ClassName() {
      // constructor body
   }
}
```

Types of Constructors in Java

1. Default Constructor

- A constructor with **no parameters**
- Provided automatically by Java if no constructor is written

```
class Student {
}
```

2. No-Argument Constructor (User-Defined)

- Created by programmer
- Does not accept parameters

```
class Student {
   Student() {
      System.out.println("Student Created");
   }
}
```

3. Parameterized Constructor

- Accepts parameters
- Used to initialize object with specific values

```
class Student {
   String name;
Student(String n) {
      name = n;
   }
}
```

Student s1 = new Student();

Constructor vs Method

Feature	Constructor	Method
Name	Same as class	Any valid name
Return type	No return type	Must have return type
Purpose	Initialize object	Perform operations
Called	Automatically	Called manually

Example Program

```
class Student {
   String name;
   int age;
Student(String n, int a) {
      name = n;
      age = a;
   }
void display() {
      System.out.println(name + " " + age);
   }
public static void main(String[] args) {
      Student s1 = new Student("Rahul", 20);
      s1.display();
   }
}
```

Output:

Rahul 20

Important Points (Exam-Oriented)

- Constructor initializes objects
- Constructor name must match class name
- No return type
- Called automatically when object is created
- Can be overloaded
- Cannot be inherited
- Used with new keyword

Polymorphism in Java

Polymorphism means **"many forms"**.
In Java, polymorphism allows a single method, object, or operator to behave differently in different situations.
In simple words:
One name → Multiple behaviors

Types of Polymorphism in Java

There are two main types:

1. **Compile-Time Polymorphism (Static Polymorphism)**
2. **Run-Time Polymorphism (Dynamic Polymorphism)**

1. Compile-Time Polymorphism (Method Overloading)

This occurs when multiple methods have the **same name but different parameters** in the same class.

Example:

```
class MathOperation {
int add(int a, int b) {
      return a + b;
   }
int add(int a, int b, int c) {
      return a + b + c;
   }
}
class Main {
   public static void main(String[] args) {
      MathOperation obj = new MathOperation();
System.out.println(obj.add(5, 10));
      System.out.println(obj.add(5, 10, 15));
   }
}
```

Output:

15
30
This is called **Method Overloading**.

2. Run-Time Polymorphism (Method Overriding)

This occurs when a **child class provides its own implementation** of a method defined in the parent class.
Achieved using **Inheritance**.

Example:

```
class Animal {
   void sound() {
      System.out.println("Animal makes sound");
   }
}
class Dog extends Animal {
   void sound() {
      System.out.println("Dog barks");
   }
}
class Main {
   public static void main(String[] args) {
      Animal obj = new Dog();
      obj.sound();
   }
}
```

Output:

Dog barks
This is called **Method Overriding**.

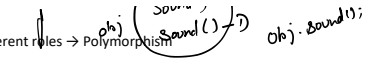
Real-Life Example

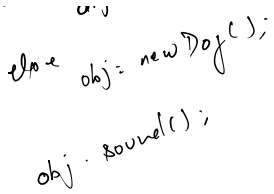
Person → can behave as:

- Teacher
- Student

- Employee

Same person → different roles → Polymorphism





Method Overloading vs Method Overriding

Feature	Method Overloading	Method Overriding
Occurs in	Same class	Parent and Child class
Parameters	Must be different	Must be same
Return type	Can be same or different	Must be same
Inheritance required	No	Yes
Compile time / Runtime	Compile Time	Runtime

Advantages of Polymorphism

- Code reusability
- Flexibility
- Improves readability
- Supports dynamic method binding

Important Exam Points

- Polymorphism means many forms
- Achieved using:
  - Method Overloading
  - Method Overriding
- Overloading → Compile time polymorphism
- Overriding → Runtime polymorphism
- Runtime polymorphism uses inheritance

Abstraction in Java

Abstraction means **hiding implementation details and showing only essential features to the user**.

In simple words:

What to do → shown

How to do → hidden

Real-Life Example

Example: **ATM Machine**

You can:

- Withdraw money
- Check balance

But you don't know:

- How the ATM processes internally

This is **abstraction**.

How Abstraction is Achieved in Java

Abstraction is achieved using:

1. **Abstract Class**
2. **Interface**

1. Abstract Class

An **abstract class** is a class declared using the abstract keyword.

It can have:

- Abstract methods (without body)
- Normal methods (with body)

Syntax:

```
abstract class ClassName {
   abstract void methodName();
}
```

Example:

```
abstract class Animal {
   abstract void sound();
}
class Dog extends Animal {
   void sound() {
      System.out.println("Dog barks");
   }
}
class Main {
   public static void main(String[] args) {
      Dog d = new Dog();
      d.sound();
   }
}
```

Important Points about Abstract Class

Important Points about Abstract Class

- Declared using abstract keyword
- Cannot create object directly
- Can have abstract and normal methods
- Must be inherited

2. Interface

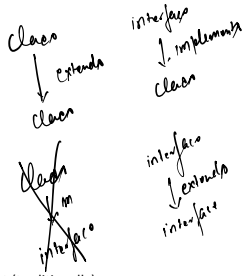
An **interface** is used to achieve complete abstraction.

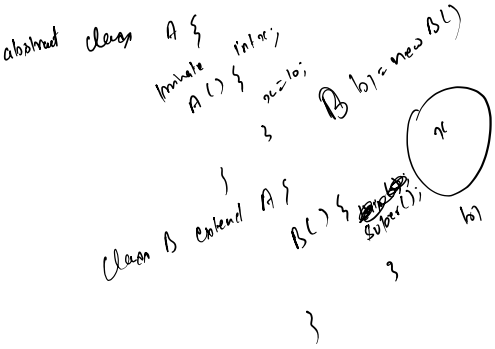
Syntax:

```
interface Shape {
   void draw();
}
```

Example:

```
interface Shape {
   void draw();
}
class Circle implements Shape {
   public void draw() {
      System.out.println("Drawing Circle");
   }
}
```





Abstract Class vs Interface

Feature	Abstract Class	Interface
Methods	Abstract + Normal	Only abstract (traditionally)
Keyword	abstract class	interface
Inheritance	extends	implements
Multiple inheritance	No	Yes

Advantages of Abstraction

- Hides internal implementation
- Improves security
- Reduces complexity
- Improves maintainability
- Supports loose coupling

Important Exam Definition (2 Marks)

Abstraction is the process of hiding implementation details and showing only essential features to the user. It is achieved using abstract classes and interfaces in Java.